

HAT-P-19b

R-1586

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-19_220806 · created 2026-08-02T00:31:45+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459797.8541 +/- 0.0045 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.099 +/- 0.053
Transit depth (Rp/Rs)^2	0.99 +/- 1.06 [%]
Orbital Inclination (inc)	85.62 +/- 1.58
Airmass coefficient 1 (a1)	0.99 +/- 0.58
Airmass coefficient 2 (a2)	0.19 +/- 0.23
Scatter in the residuals of the lightcurve fit is	64.88 %
Best Comparison Star	None
Optimal Aperture	1.46
Optimal Annulus	7.25
Transit Duration (day)	0.042 +/- 0.042

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.099 ± 0.053 vs 0.1418 (deviation 0.8σ)
Duration fit vs published	0.042 d vs 0.1196 d (deviation 1.83σ)
Mid-time O–C	10.7 min
Depth SNR	1.9
Residual scatter	64.88 %

Light curve

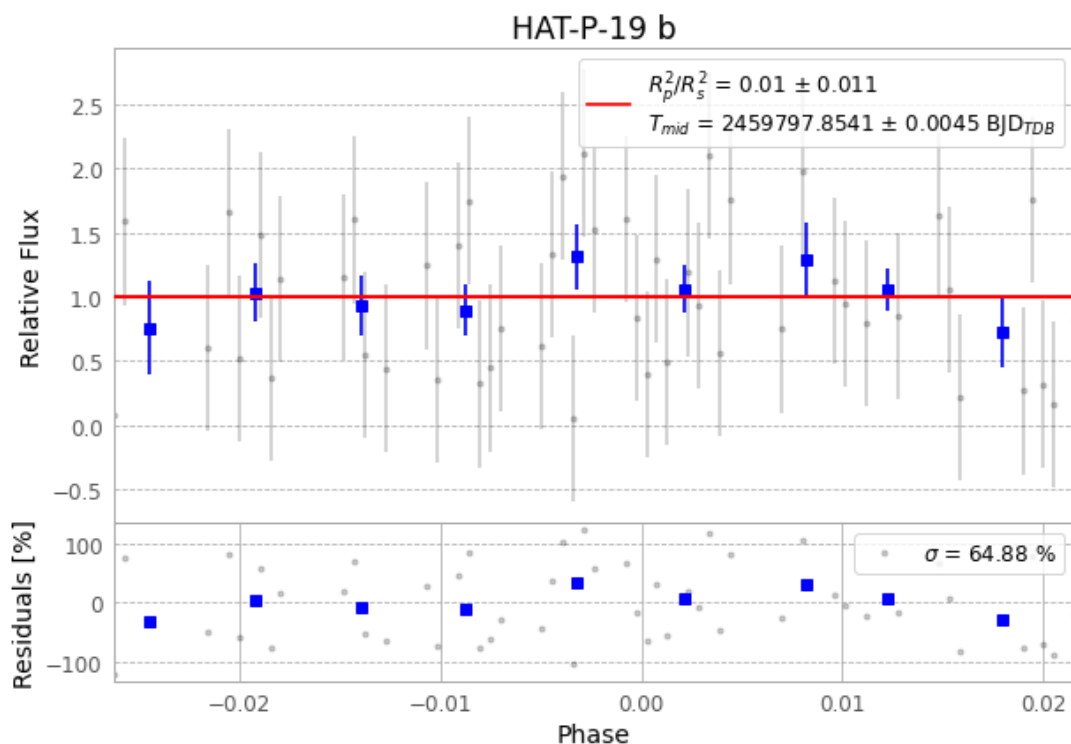


Figure 1 — FinalLightCurve_HAT-P-19 b_2022-08-05.png (SHA-256 47776df24858dfe7...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [364, 296], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 37d01a42b4b53fd4...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

97 FITS frames (64 MB), HATP-19220806055413.FITS → HATP-19220806104213.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-19-220806.log (5a3b7ad8b86f...), FinalLightCurve_HAT-P-19 b_2022-08-05.png (47776df24858...), FinalLightCurve_HAT-P-19 b_2022-08-05.csv (1d35e16a790c...)

Compute resources

Wall time 165.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 00:31Z.